FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Intrauterine Fetal Death

Tutorial | Week 2 · Labour, delivery & the puerperium | Obstetrics | Dr Nanzira

Intrauterine fetal death (IUFD) is the death of the fetus in the uterus after the period of abortion (28 weeks or more in Uganda). Confirm it by ultrasound, check the mother for pre-eclampsia, abruption, sepsis and DIC, then support her to deliver vaginally (usually by induction with mifepristone and misoprostol at a dose matched to the gestation). Investigate the cause, care for the bereaved family, and plan to prevent a recurrence.

Must know before the ward round

1. Diagnosis is made by real-time ultrasound showing no fetal cardiac activity, ideally confirmed by a second operator. Absent heart sounds on Pinard or Doppler alone are not enough.
2. First assess the mother: BP and urine protein (pre-eclampsia), abdominal pain and a hard uterus (abruption), fever (chorioamnionitis), and FBC with platelets, clotting and fibrinogen (DIC).
3. DIC screen: platelets, PT/APTT and fibrinogen where available; everywhere, a bedside clotting test (2 mL in a plain glass tube, no clot by 7 minutes = coagulopathy). Fibrinogen under 2 g/L is a danger sign.
4. DIC risk with a retained dead fetus is about 10% within 4 weeks and up to about 30% after, and higher and faster with abruption or sepsis.
5. Deliver urgently if there is sepsis, pre-eclampsia, abruption, ruptured membranes or coagulopathy. Otherwise she may choose expectant care (about 85–90% labour within 2–3 weeks), with clotting checked twice weekly after 48 hours.
6. Vaginal birth is recommended for most. Caesarean section for a dead baby adds maternal risk now and in every later pregnancy.
7. Induction: mifepristone 200 mg orally, then misoprostol after 24–48 hours. Reduce the misoprostol dose as gestation rises: 14–28 weeks, 400 µg (WHO) or 200 µg (FIGO) every 4–6 hours; after 28 weeks, 25 µg vaginally every 6 hours or 25 µg orally every 2 hours (FIGO). Use lower doses with a uterine scar.
8. Wait at least 4 hours after the last misoprostol dose before starting oxytocin. Never give 200 µg misoprostol in the third trimester.
9. Fresh stillbirth (skin intact) usually means death in labour, which is a quality-of-care signal. Macerated stillbirth (skin peeling) means death before labour.
10. Take the Kleihauer and syphilis test before delivery; the examination of the baby and placenta (plus histology) is the most informative cause-finding step in Uganda.
11. After birth: give anti-D within 72 h if Rh-negative, suppress lactation (cabergoline 1 mg once within 24 h, not in pre-eclampsia), and provide respectful bereavement care and an MPDSR review.
12. Follow-up at about 6 weeks: results, OGTT and antiphospholipid antibodies if indicated, early booking and aspirin 75–150 mg from 12 weeks where placental disease was the cause.

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1. Definition

Intrauterine fetal death (IUFD) is the death of a fetus in the uterus after the gestation at which a loss is called an abortion. In **Uganda that is from 28 weeks**. WHO international comparisons use 28 weeks, and ICD uses 22 weeks or 500 g. A baby born with no signs of life after IUFD or intrapartum death is a **stillbirth**.

Term	Meaning
Intrauterine fetal death (IUFD) / fetal demise	Fetal death in utero; the term is used for antepartum death
Stillbirth (WHO, for international comparison)	Baby born with no signs of life at **28 weeks or more** (or 1,000 g or more). Losses from 22–27 weeks are "early stillbirths" in some systems
Uganda practice	Loss **before 28 weeks = abortion**; **28 weeks or more = stillbirth** (recorded and reviewed in MPDSR)
Antepartum stillbirth	Death before labour began. Usually **macerated**
Intrapartum stillbirth	Death during labour. Usually **fresh**
Fresh stillbirth (FSB)	Skin intact, no peeling. Death probably within about **12 hours** before birth, usually intrapartum
Macerated stillbirth (MSB)	**Skin slippage and peeling, soft collapsing skull, red-brown discolouration.** Death more than about **12–24 hours** before birth
Late IUFD (RCOG)	Fetal death from 24 weeks onwards

The mother is now the patient. **The dead fetus is not an emergency, but its causes (abruption, pre-eclampsia, sepsis, uterine rupture) and its complications (DIC, infection) can kill her.**

The core tasks:

1. **Confirm** the death.
2. **Break the news** well.
3. **Assess and stabilise** the mother.
4. **Plan delivery**, preferably vaginal.
5. **Find the cause.**
6. **Care for the bereaved family.**
7. **Prevent recurrence.**

2. Epidemiology

- The global burden is about **2 million stillbirths a year**, mostly in sub-Saharan Africa and South Asia. **About half of African stillbirths happen during labour**, so they are largely preventable with good intrapartum care.
- Uganda's stillbirth rate is estimated at roughly **15–20 per 1,000 total births** (UN IGME estimates). That is tens of thousands of stillbirths every year, several times the rate in high-income countries.
- **A large share of Ugandan stillbirths are fresh (intrapartum):**
  - delayed recognition of obstructed or prolonged labour;

    - poor fetal monitoring (the partograph or Labour Care Guide is not filled in);
    - delays in referral and in CS decisions;
    - uterine rupture.
- The **"three delays" model** explains many of them:
    - **delay in deciding to seek care**;
    - **delay in reaching care**;
    - **delay in receiving adequate care**.
- **Health-system realities:**
    - **ultrasound is at HC IV and above**, with variable sonographer skill;
    - **clotting tests (PT/APTT, fibrinogen) are only at hospitals**, so use the **bedside clotting test**;
    - **mifepristone may not be stocked** in public facilities; misoprostol is widely available;
    - **postmortem examination is rarely available or accepted**;
    - bereaved mothers are often nursed in open wards beside mothers with live babies.

> **UGANDA CONTEXT**
> Every stillbirth should be **notified and reviewed** under Uganda's **Maternal and Perinatal Death Surveillance and Response (MPDSR)** system. Classify it (fresh vs macerated), identify the likely cause and the **avoidable factors / delays**, and act on them. Clerks can help by documenting carefully: FHR findings at admission, partograph entries, and timings.

3. Aetiology and Risk Factors

Macerated stillbirths point to antenatal problems:

- **hypertension and pre-eclampsia**;
- **malaria in pregnancy**;
- **syphilis**;
- **abruption**;
- diabetes;
- FGR (fetal growth restriction) from placental insufficiency;
- infection;
- congenital anomalies;
- post-term pregnancy;
- sickle cell disease;
- anaemia.

Fresh stillbirths point to labour: obstructed or prolonged labour, poor fetal monitoring, oxytocin misuse (excessive contractions), referral and CS delays, uterine rupture, cord prolapse.

Causes grouped by mechanism

Mechanism	Examples
Placental insufficiency	Pre-eclampsia, chronic hypertension, SLE / antiphospholipid syndrome, smoking, malaria (placental sequestration of *P. falciparum*), sickle cell disease, post-term
Acute placental separation	**Abruption** (hypertension, trauma, cocaine)
Cord accidents	Cord prolapse, true knots, tight entanglement, velamentous cord

Mechanism	Examples
Intrapartum asphyxia	Obstructed or prolonged labour, excessive contractions (oxytocin misuse), **uterine rupture**
Infection	Syphilis, chorioamnionitis (e.g. GBS, *E. coli*, *Listeria*), malaria, CMV, parvovirus B19 (fetal anaemia, hydrops)
Maternal metabolic	**Diabetes**, **obstetric cholestasis**, thyroid disease
Fetal	Chromosomal and structural anomalies, hydrops (Rh isoimmunisation, parvovirus), twin–twin transfusion
Unexplained	Up to about a third after full investigation; often subtle placental pathology

Risk factors for DIC: retention of the dead fetus for 4 weeks or more, abruption, sepsis.

Risk factors for recurrence: placental disease, diabetes and APS (higher); cord accidents (low).

4. Pathophysiology

Why do fetuses die? The final common pathway is hypoxia

Most causes end in **fetal hypoxia and acidosis**, either **chronically** (the placenta fails slowly) or **acutely** (the oxygen supply is cut off suddenly).

Mechanism	Why the fetus dies
Placental insufficiency	Poor spiral artery remodelling and placental infarcts reduce oxygen and nutrient transfer. The result is FGR, then decompensation and death
Acute placental separation	Loss of exchange surface → acute hypoxia. A 50% abruption is usually fatal to the fetus
Cord accidents	Blocked umbilical flow
Intrapartum asphyxia	Repeated contractions without recovery, or sudden loss of the placental bed
Infection	Direct fetal infection, placentitis, fetal anaemia and heart failure
Maternal metabolic	Diabetes: fetal hyperglycaemia → hyperinsulinaemia → increased O_2 demand, acidosis. Obstetric cholestasis: bile acids → fetal arrhythmia
Fetal	Incompatible with life, or heart failure

Why the retained dead fetus causes DIC

- After death, the placenta and decidua gradually release **thromboplastin (tissue factor)** into the maternal circulation.
- This triggers **low-grade consumption of clotting factors and platelets**, with **falling fibrinogen**: a chronic **consumptive coagulopathy (DIC)**.
- **Timing (RCOG):**
  - **around 10% of women develop DIC within 4 weeks** of fetal death;
  - **up to about 30% after 4 weeks**;
  - the risk is **higher still with abruption or sepsis**, where DIC can be acute and severe.
- DIC matters at delivery because the placental bed must stop bleeding by **contraction plus clotting**. With low fibrinogen and platelets, **PPH can be torrential**.

Why the uterus needs less prostaglandin as gestation advances

- The **density of oxytocin and prostaglandin receptors in the myometrium increases** as pregnancy progresses, and the cervix softens.
- A dose of misoprostol that is safe at 16 weeks (e.g. 400 µg) can cause **hyperstimulation and uterine rupture** at 36 weeks. That is why misoprostol doses **fall as gestation rises** and fall further with a uterine scar.
- **Mifepristone** blocks progesterone, the hormone that keeps the uterus quiet. It **primes** the uterus and cervix and shortens the time from induction to delivery.

Cause: placental insufficiency, abruption, cord, infection, labour asphyxia	FHR	fundal
falling		
↓		
Fetal hypoxia and acidosis		
↓		
Fetal death in utero		
↓		
D	Retained fetus: placenta and decidua release tissue factor	Skin and tissue autolysis: maceration, Spalding sign
Ruptured membranes or long interval: ascending infection, sepsis		
↓		
F		
↓		
DIC: oozing, non-clotting blood		
↓		
Torrential PPH at delivery		

Feature links

- Fetal hypoxia and death → absent fetal movements → mother presents with "the baby has stopped moving".
- Chronic placental insufficiency → FGR → fundal height less than dates; reducing movements before death.
- No cardiac activity → absent FHS on Pinard or Doppler (suggestive only) → absent cardiac activity on real-time ultrasound (diagnostic).
- Death days ago → autolysis → maceration, soft collapsing skull, overlapping bones on VE and imaging (Spalding).
- Tissue factor release → consumptive coagulopathy → low platelets, prolonged PT/APTT, fibrinogen under 2 g/L, bleeding from drip sites and gums, haematuria, bedside clotting time over 7 minutes.
- Abruption → retroplacental bleeding → constant pain, woody-hard tender uterus, vaginal bleeding, shock, acute DIC.
- Pre-eclampsia (cause) → high BP, proteinuria, headache, epigastric pain; HELLP → low platelets, raised transaminases.
- Ruptured membranes or a long interval → ascending infection → fever, tachycardia, offensive liquor.
- Uterine rupture → scar pain, fetal parts easily felt, shock, labour that stops suddenly.
- Raised receptor density in late pregnancy → small misoprostol doses suffice → high doses cause hyperstimulation and rupture.

Radiological and ultrasound signs of fetal death (useful for vivas)

- **Ultrasound:**
  - absent cardiac activity (the diagnostic sign);
  - later, overlapping skull bones, scalp oedema, collapsed or distorted body, absent movement.
- **Plain X-ray** (historical, but still asked):
  - **Spalding's sign:** overlapping of the skull bones, about **5–7 days** after death;
  - **Robert's sign:** gas in the fetal great vessels / heart, from about 12 hours;
  - **Deuel's halo sign:** a halo around the skull from subcutaneous oedema;
  - **hyperflexion or crumpling of the spine.**

5. Classification / Staging

By timing of death

Type	Skin	Timing	Points to
Fresh stillbirth	Intact, no peeling	Within about 12 h of birth	Intrapartum death: quality of labour care
Macerated stillbirth	Slippage and peeling, soft skull, red-brown	More than about 12–24 h	Antepartum cause

Maceration grades (roughly how long ago death occurred)

Grade	Features	Approximate time since death
0	Reddish skin, no peeling	Under 8 hours
I	Skin slippage, early peeling	Over 8 hours
II	Extensive peeling, effusions (fluid in body cavities)	Days
III	Liver yellow-brown, turbid effusions, possible mummification	Over 1 week

By gestation (drives the induction regimen): 14–28 weeks (managed on the late-abortion pathway in Uganda), 27–28 weeks, over 28 weeks; and **scarred vs unscarred uterus** (section 9).

6. Clinical Features

History

The usual story on a Ugandan ward: a woman arrives at the labour suite or ANC clinic saying **"the baby has stopped moving"**, and the midwife cannot find the fetal heart.

Ask	Why it matters (what changes the plan)
When did you last feel the baby move? Were movements reducing before that?	Dates the death (longer retention = more DIC risk; maceration grade); reducing movements suggest chronic placental insufficiency
Gestational age: LMP, early scan, fundal height at booking	Decides the **misoprostol dose** and whether this is a stillbirth (28 weeks or more) or managed as a late abortion
Abdominal pain, vaginal bleeding, a hard uterus	**Abruption**, and with it the risk of DIC and shock: resuscitation, prompt delivery (ARM + oxytocin)
Headache, visual disturbance, epigastric pain, swelling	**Pre-eclampsia / HELLP**: $MgSO_4$, antihypertensives, and **no cabergoline** later

Ask	Why it matters (what changes the plan)
Fever, offensive discharge, leaking liquor (how long?)	**Chorioamnionitis / sepsis**: antibiotics and urgent delivery; ruptured membranes rule out expectant care
Previous CS, myomectomy or uterine surgery; how many scars; classical?	**A scar changes the induction plan**: lower misoprostol doses or Foley + oxytocin; hospital with theatre; CS considered only for classical or more than two scars
Itching (palms and soles)	**Obstetric cholestasis**
Excessive thirst, known diabetes, big babies before	Diabetes
Malaria episodes, IPTp doses, bed-net use	Malaria in pregnancy
ANC card: syphilis, HIV, Hb, blood group, BP trend, IPTp doses	Infectious and hypertensive causes; anti-D; eMTCT
Trauma, domestic violence	Abruption, and a safeguarding need
Drugs, herbs (local oxytocics), alcohol, smoking	Hyperstimulation or toxic causes
Previous obstetric history: stillbirths, FGR, pre-eclampsia	Recurrent causes; APS; next-pregnancy plan
Blood group / Rhesus	Anti-D; isoimmunisation as a cause
Social: who is with her, support at home, partner, religious and cultural wishes about the baby	Breaking the news, seeing the baby, bereavement care and burial plans

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), adapted for the obstetric patient. If she is shocked, bleeding, fitting or febrile, the first-5-minutes assessment (section 9) comes first.

Introduction, consent and preparation

- **Introduction and identity**: introduce yourself; confirm her name, age and gestation.
- **Explanation and consent**: explain each step; gain consent; **offer a chaperone** and ensure privacy. Do not announce the death at this stage (section 9, Communication).
- **Hand hygiene** before and after.
- **Position and exposure**: semi-recumbent with a left lateral tilt (avoids aortocaval compression); expose the abdomen from the xiphisternum to the symphysis; cover the legs.

General inspection and vital signs

Sign	What it means
Pallor	Anaemia, haemorrhage (abruption); baseline for PPH
Distress, pain, guarding	Abruption, rupture
Oedema (face, hands)	Pre-eclampsia
Jaundice	HELLP, cholestasis, haemolysis, sepsis
Bruising, oozing from puncture sites or gums, haematuria	**DIC**

Vital signs: pulse, **BP**, **temperature**, respiratory rate, SpO_2; shock index (pulse ÷ systolic BP: **0.9 or more abnormal, 1.4 or more critical**). **Urine dipstick**: **protein** (pre-eclampsia), glucose (diabetes), nitrites (UTI).

Hands, face and neck

- **Hands**: pallor, capillary refill, cold peripheries (shock); scratch marks (cholestasis).
- **Face**: conjunctival pallor, scleral jaundice, facial oedema; bleeding gums (DIC).
- **Neck**: brief; thyroid if thyroid disease is suspected.
- **Reflexes** (brisk with clonus in severe pre-eclampsia) if hypertensive.

Obstetric abdominal examination

Step	Finding	What it means
Inspection	Distension, scars (previous CS, myomectomy), fetal movements absent	A scar changes the induction plan
Fundal height	Often less than dates if death was days ago or FGR	Chronic placental insufficiency, death some days ago
Lie	Longitudinal, transverse	Transverse lie in labour may need CS
Presentation	Cephalic, breech	Plans delivery
Engagement (fifths palpable)	Fifths above the brim	Progress, obstruction
Palpation for tone	**Tenderness, woody-hard uterus**	**Abruption**
Palpation of the scar	**Scar tenderness**; fetal parts unusually easy to feel	**Uterine rupture**
Auscultation of the fetal heart (Pinard or Doppler)	Absent	**Suggestive only**; confirm by ultrasound

Speculum and vaginal examination

Only if in labour, bleeding or membranes ruptured; **not** if praevia is possible until a scan has excluded it.

- liquor colour and smell; bleeding; **cord prolapse**;
- cervical dilation and the **Bishop score** (chooses misoprostol vs Foley vs ARM + oxytocin);
- **signs of maceration**: a soft, collapsing skull with overlapping bones.

Completing the examination

- **Point-of-care ultrasound** if trained (section 8), and arrange formal confirmation.
- **Thank** her, cover her, **wash hands**.
- **Summarise** findings (model presentation in section 8).

Document: BP, protein, temperature, fundal height, uterine tone and tenderness, scar status, Bishop score, and signs of bleeding.

Red flags

RED FLAG

- **BP 160/110 or more, proteinuria, headache or epigastric pain.** Severe pre-eclampsia / HELLP. Give $MgSO_4$ and antihypertensives and deliver.
- **Constant abdominal pain, tense tender uterus, bleeding, shock. Abruption**: resuscitate and check for DIC.
- **Fever, tachycardia, offensive liquor. Chorioamnionitis / sepsis**: antibiotics and delivery.
- **Bleeding from drip sites, haematuria, bedside clotting time over 7 minutes, low platelets or fibrinogen. DIC**.
- **Previous CS with scar tenderness, or fetal parts easily felt. Uterine rupture**: this is not a case for induction; laparotomy.
- **Death undiagnosed or retained for 4 weeks or more.** Higher DIC risk.

Thinking like a clinician

- **Confirm before you counsel.** A Pinard or hand-held Doppler that "cannot find the heart" can be wrong (obesity, polyhydramnios, operator error, a maternal pulse mistaken for the fetal heart rate). **Scan, and if possible have a second person confirm.** Only then break the news.
- **Fresh or macerated? The answer points you to the cause:**
  - **Fresh** means intrapartum death. Ask **what went wrong in labour**: the partograph or Labour Care Guide, FHR monitoring, referral delays, oxytocin use, obstructed labour or rupture.
  - **Macerated** means antenatal causes: hypertension, infection, diabetes, FGR, anomalies.
- **Is the mother safe to wait?** This is the key management fork:
  - **Urgent delivery:** pre-eclampsia, abruption, sepsis or ruptured membranes, DIC, maternal request.
  - **Otherwise:** expectant care is allowed. Most women prefer to deliver soon.
- **"Can we just do a caesarean?"** Families often ask. Explain gently that **a CS carries surgical risk now and scar risks in every future pregnancy** (uterine rupture, placenta accreta), with **no benefit to the baby**. In Uganda, a woman may not reach a CS-capable facility in her next labour, so **avoiding an unnecessary scar is lifesaving**.
- **Know the scar.** With a previous CS, use **lower misoprostol doses, a Foley catheter or oxytocin**, monitor closely for scar pain and bleeding, and deliver in a hospital with theatre. The risk of uterine rupture with induction in a scarred uterus is small but real.
- **A "normal" fibrinogen may not be normal.** Pregnancy raises it to 4–6 g/L, so 2.5 g/L is already falling.
- **Common student mistakes:**
  - Announcing fetal death after a Doppler alone.
  - Forgetting to check BP and urine.
  - Using a term-induction misoprostol dose at 20 weeks, or a 20-week dose at 38 weeks.
  - Giving oxytocin within 4 hours of misoprostol.
  - Forgetting the placenta and baby examination.
  - Forgetting anti-D and lactation suppression.
  - Talking about "the fetus" instead of **using the baby's name** if the parents have chosen one.

7. Differential Diagnoses

When fetal heart sounds cannot be heard:

Differential	Distinguishing clinical feature	Distinguishing investigation
IUFD	Absent movements, fundal height static or reduced	**Ultrasound: no cardiac activity**
Technical failure (obesity, polyhydramnios, posterior position, anterior placenta, maternal pulse counted)	Mother still feels movements	Ultrasound shows a heartbeat
Pseudocyesis / wrong dates	No fetus	Negative hCG; ultrasound
Molar pregnancy	Uterus large for dates, no fetal parts, early pre-eclampsia, hyperemesis	Ultrasound snowstorm; very high hCG
Abdominal pregnancy with fetal death	Fetal parts superficial, uterus separate	Ultrasound

Differential causes of death (keep in mind while working it up): abruption, pre-eclampsia, infection (syphilis, malaria, chorioamnionitis), diabetes, cholestasis, FGR / placental insufficiency, cord accident, anomaly, isoimmunisation, uterine rupture, unexplained.

8. Clinical Workup

Two parallel tracks: **confirm the death** (gently, without announcing it prematurely) and **make sure the mother is safe** (pre-eclampsia, abruption, sepsis, DIC, uterine rupture). Then **find the cause**.

Bedside

Test	Why / what it tells you	Expected finding	Available at
Point-of-care ultrasound	Confirms death	**Absent cardiac activity** over at least a few minutes of real-time scanning, ideally confirmed by a second operator. Also: presentation, placental site (praevia?), retroplacental clot, liquor volume, overlapping skull bones (Spalding), scalp oedema	HC IV and above (trained user)
Bedside clotting test	DIC screen everywhere	2 mL of blood in a plain glass tube held in the hand: **no clot by 7 minutes**, or a clot that breaks down = coagulopathy. Repeat before and during induction if abnormal or at risk	All levels with a glass tube
Urine dipstick	Pre-eclampsia, diabetes, UTI	Protein, glucose, nitrites	All levels
HemoCue / bedside Hb	Anaemia; baseline for PPH	Low Hb	HC III and above
Random blood sugar	Diabetes as a cause	A very high RBS points to the cause	HC III and above (variable)
mRDT	Malaria	Positive: treat; a possible cause	All levels
Rapid syphilis and HIV tests	Infectious causes; eMTCT	Reactive: start **benzathine penicillin** on a reactive syphilis test	All levels
Partograph / Labour Care Guide review	Fresh stillbirth: where did monitoring or action fail?	Missed alerts, delays	Feeds the MPDSR review

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC with platelets	Everyone	Low Hb; **platelets under 100** (DIC, HELLP); raised WBC (sepsis)	Transfusion; blood products; urgent delivery	HC IV and above
Group, Rhesus, crossmatch, antibody screen	Everyone	Rh-negative; red cell antibodies	Blood ready for PPH; anti-D; isoimmunisation as a cause (hydrops)	HC IV and above
Creatinine, LFTs, urate	Hypertension, sepsis or suspected HELLP	Raised creatinine or transaminases	HELLP/pre-eclampsia management; gentamicin dosing	Some HC IVs; hospitals
Syphilis serology (RPR/VDRL + TPHA / rapid test)	Everyone	Positive	Common, treatable cause in Uganda; benzathine penicillin for her and her partner	RPR at hospitals; rapid test everywhere
HIV test	Everyone	Positive	Care and linkage	All levels
Random blood sugar / HbA1c (± OGTT at 6 weeks postnatally)	Everyone	Hyperglycaemia	Diabetes (overt or gestational)	HC IV and above (HbA1c at hospitals)
Sickle cell test (if unknown)	Status unknown	HbSS / HbSC	Placental vaso-occlusion as a cause	Hospitals

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Formal obstetric ultrasound	Everyone to confirm; second operator where possible	No cardiac activity; Spalding's sign; placental position; abruption clot (often not seen)	Confirms the diagnosis; praevia changes the mode of delivery	HC IV and above
Plain X-ray	Historical; not needed	Spalding, Robert, Deuel signs (section 4)	None today	Hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
PT/APTT, fibrinogen	Everyone, where available; **repeat twice weekly** if expectant beyond 48 h, and before and during induction if abnormal	Prolonged PT/APTT; **fibrinogen under 2 g/L**	DIC: correct with fresh whole blood, FFP, platelets or cryoprecipitate before and during delivery; avoid IM injections and spinal/epidural	Regional/national referral
Kleihauer–Betke	Rh-negative women and any suspected fetomaternal haemorrhage; **before delivery**	Fetal red cells in the maternal circulation	Large **fetomaternal haemorrhage** as a cause; sizes the **anti-D** dose	Referral hospitals only
Blood cultures, HVS	Febrile women	Organism	Antibiotic choice	Referral hospitals

Test	When	Expected / abnormal finding	How it changes management	Available at
Bile acids and LFTs	Itching	Raised	Obstetric cholestasis	Referral hospitals
Thyroid function	If indicated	Abnormal TSH	Thyroid disease	Referral hospitals
Antiphospholipid antibodies (lupus anticoagulant, anticardiolipin, anti-β2-GP1), about 6 weeks after delivery	Placental disease, recurrent loss	Positive	APS, which is treatable next time with aspirin and heparin	NRH, private labs
Viral serology (CMV, toxoplasma, parvovirus B19, rubella)	Hydrops or FGR	Recent infection	Infectious cause	Referral hospitals
Examination of the baby: weight, sex, maceration grade, **external anomalies**, photos (with consent), cord (knots, length, vessels)	Every stillbirth	Anomaly, FGR, cord cause	Available everywhere, and often the best "postmortem" in Uganda	All levels
Placenta: weight, **retroplacental clot**, infarcts, membranes, cord insertion; **histology** + swabs	Every stillbirth	Abruption, infarction, chorioamnionitis, malaria pigment	**The most informative single investigation**	Gross exam everywhere; histology at referral hospitals
Postmortem (full or limited, with consent), **karyotype / genetics** (placenta, cord or skin)	Anomalies, unexplained	Anomalies, genetic causes	Offer where available. Many families decline for cultural or religious reasons, and that is their right	NRH, rarely

Putting it together

Model one-line summary

> "A 27-year-old G2P1 at 32 weeks with absent fetal movements for 3 days; BP 150/100, **protein 2+**, afebrile, uterus soft and non-tender, no scar; **IUFD confirmed on ultrasound by two operators**, cephalic, fundal placenta; platelets 88 × 10^9/L, **bedside clotting time 9 minutes**. **Working diagnosis: IUFD at 32 weeks with pre-eclampsia and evolving coagulopathy (DIC).**"

Problem list

1. IUFD at 32 weeks (confirmed).
2. Pre-eclampsia (BP 150/100, proteinuria): control BP; $MgSO_4$ if severe features develop; **no cabergoline**.
3. Coagulopathy: correct before and during delivery.
4. Cause to find (likely placental insufficiency from pre-eclampsia; syphilis, malaria, diabetes to exclude).
5. Bereavement, anti-D, contraception, follow-up.

Plan: crossmatch and fresh whole blood/FFP; induce **now** (not expectant) with **misoprostol 25 µg vaginally every 6 hours** (or 25 µg orally every 2 hours), oxytocin later (at least 4 hours after the last misoprostol); AMTSL with TXA ready; avoid IM injections and spinal while clotting is abnormal.

Model case presentation (ward round)

> "Mrs A, a 29-year-old G3P2, both previous babies alive and delivered vaginally, at 34 weeks by an early scan, presented with **absent fetal movements for 2 days**. She is **normotensive, BP 118/72, urine protein negative, afebrile**, with no abdominal pain or bleeding. Fundal height 31 cm; no FHS on Doppler. **Ultrasound, confirmed by 2 sonographers, shows no fetal cardiac activity with overlapping skull bones.** Hb 11.4 g/dL, platelets 210, **bedside clotting time 5 minutes**, RPR **positive**, TPHA pending, HIV negative, RBS 5.8 mmol/L, Rh-positive. **Impression: IUFD at 34 weeks, likely macerated, possibly due to syphilis.** Plan: she and her husband opted for induction today with **misoprostol 25 µg vaginally 6-hourly**, then oxytocin as needed; analgesia; examine the baby and placenta; send placenta for histology; **benzathine penicillin 2.4 MU IM weekly for 3 doses** (start now on the reactive test; do not wait for the TPHA) and partner treatment; **cabergoline 1 mg** after delivery; bereavement support; MPDSR review; follow-up at 6 weeks."

Worked example 1: confirming IUFD properly

- A midwife cannot find the FHR with a Doppler. A **maternal pulse of 110** can be mistaken for a fetal heart rate, and obesity or polyhydramnios can hide a live one.
- **Confirmation:** real-time ultrasound, looking at the **fetal heart for at least a few minutes**, with colour Doppler if available; **second operator** to confirm.
- Supporting (not diagnostic) findings: **Spalding's sign** (overlapping skull bones, about 5–7 days), scalp oedema, collapsed body, absent movement.
- Only then break the news, in private, with a warning shot and clear words.

Worked example 2: interpreting a DIC screen

Result	Normal in pregnancy	Her value	Interpretation
Platelets	150–400 × 10^9/L	**88**	Low: DIC or HELLP
Fibrinogen	Rises to **4–6 g/L**	**1.6 g/L**	**Under 2 g/L = severe coagulopathy** (a "normal-looking" 2.5 g/L is already low for pregnancy)
PT / APTT	Unchanged or slightly shorter	Prolonged	Consumption of clotting factors
Bedside clotting test	Clots within 7 min	**No clot at 9 min**	Confirms coagulopathy where no lab exists

Action: involve a senior and the blood bank; give **fresh whole blood or FFP (plus cryoprecipitate or platelets where available)**; deliver **vaginally** where possible (the fastest way to stop the tissue factor source is delivery); active third stage with **oxytocin 10 IU** and **TXA 1 g IV** if PPH; avoid IM injections, spinal or epidural; repeat the tests after delivery.

9. Management

Immediate / emergency

First 5 minutes: look for the four killers

1. **Vitals:** pulse, **BP**, temperature, respiratory rate, SpO_2; shock index.
2. **ABCDE**, looking specifically for:
   - **A/B**: airway and breathing; fitting (eclampsia).
   - **C**: **constant pain, woody-hard tender uterus, bleeding, shock** = abruption → two large-bore cannulae, crossmatch, fluids, bedside clotting test. **Previous CS with scar pain, fetal parts easily felt, shock, or labour that has stopped suddenly** = uterine rupture → resuscitate, theatre. **Bleeding from drip sites,**

gums or urine = DIC → bedside clotting test, blood (fresh whole blood is often what is available), senior now.
    - **D**: **BP 160/110 or more**, headache, visual symptoms, epigastric pain, or a fit = severe pre-eclampsia/eclampsia → **$MgSO_4$** loading dose and antihypertensive now.
    - **E**: **fever 38 °C or more, tachycardia, offensive liquor** = chorioamnionitis → **ampicillin 2 g IV + gentamicin 5 mg/kg IV** now.
3. **If stable,** she can go to a quiet, private space while you arrange the ultrasound. **Do not tell her "the baby has died" on the basis of a Pinard or Doppler.** Say you are "having difficulty hearing the heartbeat" and that a scan will check.

Urgent situations (manage the complication first)

Complication	Immediate management
Severe pre-eclampsia / eclampsia	**$MgSO_4$** (Pritchard or Zuspan regimen), antihypertensives (**nifedipine 10–20 mg orally, or hydralazine 5 mg IV**, repeated to target), then deliver (see the pre-eclampsia topic)
Abruption	Resuscitate (2 large-bore lines, crystalloid, **blood**), check clotting, **ARM + oxytocin** to speed vaginal delivery (usually rapid); correct DIC with **fresh whole blood / FFP / platelets / cryoprecipitate** where available. CS if maternal condition deteriorates and delivery is not imminent
Chorioamnionitis / sepsis	**Ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily**, **plus metronidazole 500 mg IV 8-hourly** if delivery is by CS or there is endometritis; expedite delivery
DIC	Replace blood products; **fresh whole blood** is often the most available product in Uganda; deliver vaginally where possible; **tranexamic acid 1 g IV** for PPH; avoid IM injections and regional anaesthesia
Uterine rupture	Resuscitate; **laparotomy** (repair or hysterectomy)

Definitive

Step-by-step overview

1. **Confirm** fetal death by **ultrasound** (a second operator if possible).
2. **Break the news** in private, with her partner or companion present (below).
3. **Assess the mother** (history, exam, investigations). Stabilise any **pre-eclampsia, abruption, sepsis or DIC**.
4. **Decide timing:**
    - urgent delivery if there are complications;
    - otherwise her choice of **prompt induction (recommended by most)** or **expectant management**.
5. **Decide the mode: vaginal birth** unless there is a specific indication for CS.
6. **Induce** with a regimen matched to **gestation and any uterine scar**.
7. **Labour care:** analgesia, monitor the mother (not the fetus), anticipate PPH.
8. **After birth:** examine the baby and placenta, send tests, anti-D, lactation suppression, thromboprophylaxis assessment, contraception.
9. **Bereavement care** and **MPDSR notification / review**.
10. **Follow-up at about 6 weeks:** results, cause, plan for the next pregnancy.

Communication: breaking the news (SPIKES)

- **S**etting:
    - a private room; sit down;
    - her partner or companion present;
    - phone off; enough time.

- **P**erception: "What have you noticed about the baby's movements?"
- **I**nvitation: ask how much she wants to know now.
- **K**nowledge:
  - Give a warning shot: "I am afraid I have very sad news."
  - Then say it clearly: "**Your baby has died.** I could not see a heartbeat." Do not use euphemisms ("we've lost the heartbeat").
  - Pause and allow silence.
- **E**mpathy:
  - Acknowledge her feelings; tears are normal.
  - Do not blame her.
  - Offer the support of a chaplain or religious leader if she wants.
- **S**trategy / summary:
  - Explain the next steps simply: tests, the options for birth.
  - She does not need to decide everything today.
  - Offer to explain again later; information is often not retained in shock.

Expectant management

- Suitable when she is **well, membranes are intact**, and there is **no pre-eclampsia, infection, abruption or coagulopathy**.
- About **85–90% go into spontaneous labour within 2–3 weeks**.
- Many women find waiting distressing. RCOG advises that women who delay more than 48 hours should have **clotting (and FBC) checked twice weekly**, and be told to return with fever, bleeding, pain or leaking.
- A short delay of 24–48 hours at home to prepare, with family support, is reasonable.

> **UGANDA CONTEXT**
> Long expectant management is rarely practical in Uganda: clotting tests are limited, women live far from facilities, and most families prefer early delivery. **Prompt induction in hospital is the usual approach.** Do it at a facility with **blood, theatre and oxytocin** (HC IV or hospital).

Mode of delivery

Vaginal birth for most women. Consider **caesarean section** only for:

- **major placenta praevia**;
- **transverse lie in labour / obstructed labour** that cannot be safely managed vaginally;
- **uterine rupture**;
- **previous classical CS** or more than two previous CS (individualise with a senior);
- **severe maternal compromise** where vaginal birth is not imminent (e.g. severe abruption with deteriorating mother, while correcting clotting);
- **maternal request after full counselling** (discouraged, but respect her autonomy).

> **NOTE**
> Obstructed labour with a dead fetus: in some African centres, **destructive operations** (craniotomy, decapitation, embryotomy) by an experienced obstetrician are used to avoid a CS scar (and the rupture risk it carries in the next pregnancy). You need only know that they exist and what they are for.

Induction of labour: regimens by gestation

Step 1 (if available): mifepristone 200 mg orally, once. Then start misoprostol **24–48 hours later**. This shortens induction-to-delivery time and reduces the misoprostol needed.

Step 2: misoprostol (dose falls as gestation rises):

Gestation	Regimen	Source / notes
14–28 weeks	**Misoprostol 400 µg sublingually or vaginally every 4–6 hours** (after mifepristone, or alone)	**WHO Abortion care guideline 2022** (IUFD 14–28 weeks). No fixed maximum number of doses; use clinical judgement
13–26 weeks (alternative)	**Misoprostol 200 µg** vaginally, sublingually or buccally **every 4–6 hours**	FIGO misoprostol-only regimens
18–26 weeks (alternative)	**Misoprostol 100 µg** vaginally **every 6 hours** (up to 4 doses)	Misoprostol-for-IUFD review regimen (more conservative)
27–28 weeks	**Misoprostol 100 µg** vaginally, sublingually or buccally **every 4 hours**	FIGO 2017/2023
Over 28 weeks (third trimester)	**Misoprostol 25 µg vaginally every 6 hours**, **or 25 µg orally every 2 hours** (FIGO); **25–50 µg vaginally every 4 hours**, up to 6 doses (review regimens)	Doses like those used for term labour induction. **Stop misoprostol once in active labour**
Previous CS (any gestation)	**Lower doses, no dose escalation**; consider a **Foley catheter** (16–18 F, balloon 30–60 mL) for ripening, then **oxytocin**	Senior decision; hospital with theatre; watch for scar pain, bleeding, maternal tachycardia

Step 3: oxytocin, once the cervix is favourable (**Bishop score 6 or more**) or membranes rupture, if contractions are inadequate:

- **WHO MCPC regimen: oxytocin 2.5 IU in 500 mL normal saline or Ringer's lactate** (5 mIU/mL), starting at **10 drops/min** (about 2.5 mIU/min) and **increasing by 10 drops/min every 30 minutes** until **3–4 contractions per 10 minutes, each lasting over 40 seconds**. Maximum 60 drops/min.
- If contractions are still inadequate at 60 drops/min, a higher concentration (5 IU in 500 mL) is used per protocol, except with a uterine scar.
- Many Ugandan units start with **5 IU in 500 mL**. Know your ward's protocol and **always count the contractions**.
- **Wait at least 4 hours after the last misoprostol dose** before starting oxytocin, to avoid hyperstimulation.

DRUG DOSING

Misoprostol tablets come as 200 µg. For a 25 µg dose, you must divide it. Dissolve 1 × 200 µg tablet in 200 mL of water and give **25 mL orally** (= 25 µg; discard the unused solution after 24 hours), or use purpose-made 25 µg vaginal tablets where available. **Never give 200 µg to a woman in the third trimester**: the risk of hyperstimulation and uterine rupture is real, especially with a scar.

UGANDA CONTEXT

Uganda Clinical Guidelines follow WHO/FIGO principles: misoprostol dose tailored to gestation, avoid high doses in the third trimester and in scarred uteri, and deliver at a facility able to do CS and transfuse. **Mifepristone is not consistently stocked in public facilities**, so misoprostol-alone regimens are usual. Check your hospital's protocol and write the **dose in micrograms with the route and interval** every time.

Care in labour

1. **One-to-one support.** A private room or area if possible, away from mothers with live babies. Allow a companion.
2. **Analgesia:**
   - **pethidine 50–100 mg IM** 4-hourly (or morphine), paracetamol;
   - **epidural** only if clotting and platelets are normal and she is not septic.

- Fetal side-effects are no longer a concern, so give **generous analgesia**.

3. **Monitor the mother:** pulse, BP, temperature, contractions, **vaginal bleeding**, and **scar pain** (if previous CS). **No fetal monitoring is needed.** Use a partograph for progress.
4. **Avoid early ARM** where possible (ascending infection), except in abruption, where ARM speeds delivery.
5. **Active management of the third stage: oxytocin 10 IU IM**, controlled cord traction, uterine massage. Anticipate **PPH** (DIC, atony after a prolonged induction). Have blood crossmatched. Give **tranexamic acid 1 g IV** if PPH occurs.
6. **Examine the placenta**; send it for **histology** and swabs if possible.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Mifepristone	200 mg	Oral	Once	Then misoprostol 24–48 h later	Not consistently stocked in public facilities
Misoprostol 14–28 weeks	400 µg (WHO) or 200 µg (FIGO, 13–26 weeks)	SL or vaginal (200 µg: also buccal)	Every 4–6 h	Until delivery; no fixed maximum (WHO)	Lower doses with a scar
Misoprostol 18–26 weeks (conservative)	100 µg	Vaginal	Every 6 h	Up to 4 doses	Review regimen
Misoprostol 27–28 weeks	100 µg	Vaginal, SL or buccal	Every 4 h	Until labour	FIGO 2017/2023
Misoprostol over 28 weeks	25 µg vaginal (or 25–50 µg vaginal 4-hourly, up to 6 doses) **or** 25 µg oral solution	Vaginal / oral	Vaginal 6-hourly; oral 2-hourly	Stop once in active labour	**Never 200 µg in the third trimester**; 25 µg oral = 1 × 200 µg tablet in 200 mL water, give 25 mL
Oxytocin (induction)	2.5 IU in 500 mL NS or RL; start 10 drops/min, increase by 10 every 30 min	IV infusion	Continuous	To 3–4 contractions per 10 min	Max 60 drops/min; **wait 4 h after misoprostol**; higher concentration not with a scar
Oxytocin (third stage)	10 IU	IM	Once	Once	Avoid IM if coagulopathic (use IV) Verify *route in DIC per local protocol*
Tranexamic acid	1 g	IV	Once (repeat per PPH protocol)	Once	For PPH
Magnesium sulfate	Pritchard or Zuspan regimen (see pre-eclampsia topic)	IM/IV or IV	Loading then maintenance	24 h	Reflexes, RR, urine output
Nifedipine	10–20 mg	Oral	Repeat to target	As needed	Severe hypertension
Hydralazine	5 mg	IV	Repeat to target	As needed	Severe hypertension

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Ampicillin	2 g	IV	6-hourly	Until delivered and afebrile	Chorioamnionitis
Gentamicin	5 mg/kg	IV	Once daily	As above	Renal function
Metronidazole	500 mg	IV	8-hourly	As above	Add if CS or endometritis
Pethidine (or morphine)	50–100 mg	IM	4-hourly	In labour	Avoid IM if coagulopathic
Paracetamol	1 g	Oral	6-hourly	As needed	Max 4 g/day
Benzathine penicillin	2.4 MU	IM	Weekly	3 doses	Start on a reactive rapid test; treat the partner
Anti-D immunoglobulin	UK 500 IU (20 weeks or more) + Kleihauer to size; Ugandan stock vial usually 1,500 IU	IM	Once, within 72 h	Once	Rh-negative women
Cabergoline	1 mg	Oral	Once, within 24 h of birth	Once	**Avoid in hypertension / pre-eclampsia**; bromocriptine not recommended
LMWH	Per local protocol	SC	Daily	If high VTE risk	Only if clotting normal and available
Folic acid (next pregnancy)	400 µg (5 mg if diabetes, obesity, anti-epileptics or previous NTD)	Oral	Daily	Pre-conception to 12 weeks	
Aspirin (next pregnancy, placental disease)	75–150 mg	Oral	Nightly	From 12 weeks	APS: add LMWH

Referral criteria

Level	Role	Refer up when
HC III	Recognise absent FHS / movements; check BP, urine and temperature	**Refer** every case to HC IV or hospital for scan and induction; **urgently** if she is bleeding, hypertensive or febrile
HC IV / General Hospital	Ultrasound confirmation; bedside clotting test; **misoprostol induction** (not in the scarred uterus without a doctor); manage PPH; blood transfusion; CS for rare indications	Previous CS, DIC needing blood products, severe pre-eclampsia / HELLP, need for ICU → RRH
Regional Referral Hospital	Induction with a previous CS; DIC management with blood products; severe pre-eclampsia / HELLP; full investigations (histology, serology)	ICU, karyotype, specialist counselling → NRH
National Referral Hospital	ICU; karyotype, postmortem; specialist counselling	n/a

Split between RRH and NRH adapted from the source notes, which grouped them together Verify *against UCG 2023 levels of care*.

Monitoring and follow-up

- **In labour**: pulse, BP, temperature, contractions, vaginal bleeding, scar pain; partograph for progress; no fetal monitoring.
- **Expectant care beyond 48 h**: clotting and FBC **twice weekly**; return with fever, bleeding, pain or leaking.
- **Clotting** repeated before and during induction if abnormal, and after delivery.

Follow-up (about 6 weeks)

1. Review her physical recovery, BP, mood, and lactation.
2. **Discuss results** (what was found and what was not). A "no cause found" answer must also be explained honestly. Complete any delayed tests: **OGTT**, **antiphospholipid antibodies**, thrombophilia if indicated.
3. **Plan the next pregnancy:**
    - **folic acid 400 µg daily** pre-conception (5 mg if diabetes, obesity, anti-epileptics or previous NTD);
    - **book early** (in the first trimester) with a **dating scan**;
    - **treat the cause:**
        - diabetes: tight glucose control;
        - hypertension, pre-eclampsia or placental disease: **low-dose aspirin 75–150 mg nightly from 12 weeks**;
        - APS: **aspirin + LMWH**;
        - syphilis: treat her and her partner;
    - **IPTp-SP and an ITN** (insecticide-treated net);
    - **serial growth scans** from about 28 weeks where available;
    - **fetal movement awareness**;
    - a **planned birth** at a CS-capable facility, often **around 38–39 weeks** after discussion. Do not deliver earlier without an indication.
4. **Recurrence** depends on the cause: higher with placental disease, diabetes and APS; low with cord accidents. Most women go on to have a live baby.

After the birth

Action	Details
Seeing and holding the baby	**Offer, do not force.** Many parents value it. Wrap the baby, describe what they will see (skin changes), and allow time. Offer to name the baby and create memories (footprints, a lock of hair, a name tag, a photo), respecting cultural norms
Examine and document	Weight, sex, maceration grade, anomalies, cord and placenta findings. **Fresh vs macerated**
Anti-D	If Rh-negative: **anti-D 500 IU IM** (at 20 weeks or more) **within 72 hours**. Use the **Kleihauer** to decide whether more is needed. (Many Ugandan facilities lack anti-D: document and inform her)
Lactation suppression	**Cabergoline 1 mg orally, single dose, within 24 hours of birth. Avoid in hypertension / pre-eclampsia.** Where unavailable: a firm supportive bra, cold compresses, analgesia, avoid expressing or stimulation. Warn her that milk may come in on day 2–4. **Bromocriptine is not recommended**
Thromboprophylaxis	Assess VTE risk (CS, sepsis, prolonged immobility, obesity, pre-eclampsia): early mobilisation; LMWH if high risk and available
Infection	Antibiotics if chorioamnionitis was present
Contraception	Discuss. All methods are possible. Many couples want to wait before trying again; others want to conceive soon. Advise an interval of at least about 6 months to recover physically and emotionally, but respect their choice

Action	Details
Documentation and MPDSR	Birth and death notification; perinatal death review
Burial	Respect the family's wishes: many take the baby home for burial; others ask the hospital to bury. Explain the options without pressure
Emotional care	Screen for **depression, anxiety and PTSD**; involve the partner; offer counselling and peer or religious support. Warn about grief reactions and "anniversary" distress

10. Complications

Disease-related

Complication	Notes
DIC / coagulopathy	About 10% within 4 weeks, rising later; worse with abruption or sepsis. Monitor fibrinogen and platelets; blood products
Infection / chorioamnionitis / sepsis	Especially with ruptured membranes or a long interval; antibiotics and delivery
Postpartum haemorrhage	Atony and coagulopathy; active management of the third stage, TXA, blood
Retained placenta	More common in preterm IUFD; manual removal / MVA
Psychological	Grief, depression, anxiety, PTSD, relationship strain, stigma (some communities blame the mother)
Rh isoimmunisation	If anti-D is not given
Future pregnancy	Anxiety; recurrence of the underlying cause

Treatment-related

Complication	Notes
Uterine rupture during induction	Especially with a previous CS or high misoprostol doses; respect gestation-specific doses
Hyperstimulation	Oxytocin started within 4 h of misoprostol, or high third-trimester misoprostol doses
PPH after prolonged induction	Atony; active third stage
CS scar complications in future pregnancies	If a CS was done: rupture, placenta accreta, and risk if she cannot reach a CS-capable facility next time
Haematoma at injection or spinal sites	IM injections or regional anaesthesia given with coagulopathy
Cabergoline in pre-eclampsia	Worsens hypertension: avoid

11. Prognosis

- **Maternal**: good with prompt recognition of pre-eclampsia, abruption, sepsis and DIC; most deaths come from these, not from the fetal death itself.
- **Expectant care**: about **85–90% labour spontaneously within 2–3 weeks**.
- **Recurrence** depends on the cause: higher with placental disease, diabetes and APS; low with cord accidents. **Most women go on to have a live baby.**
- **Unexplained** in up to about a third after full investigation.

- Psychological morbidity (depression, anxiety, PTSD) is common and needs follow-up.

12. Prevention

- **Antenatal (macerated stillbirth)**: detect and treat hypertension and pre-eclampsia, **syphilis**, **malaria** (IPTp-SP, ITN), diabetes, anaemia; monitor fetal growth; fetal movement awareness; avoid post-term pregnancy.
- **Intrapartum (fresh stillbirth)**: fill in the partograph or Labour Care Guide, monitor the fetal heart, recognise obstructed or prolonged labour early, use oxytocin carefully, avoid referral and CS delays (the **three delays**).
- **Next pregnancy**: folic acid, early booking with a dating scan, **aspirin 75–150 mg from 12 weeks** for placental disease, **aspirin + LMWH** for APS, tight glucose control, treat syphilis in both partners, serial growth scans, planned birth at a CS-capable facility around 38–39 weeks.
- **System**: MPDSR review of every stillbirth to identify and act on avoidable factors.

13. Special Populations

- **Previous caesarean section or uterine surgery**: lower misoprostol doses, no dose escalation; Foley catheter then oxytocin; no higher-concentration oxytocin; hospital with theatre; watch for scar pain, bleeding and maternal tachycardia. CS considered for a classical scar or more than two previous CS.
- **Rhesus-negative women**: Kleihauer before delivery; anti-D within 72 h; if anti-D is unavailable, document and inform her.
- **Pre-eclampsia / hypertension**: urgent delivery; $MgSO_4$ and antihypertensives; **no cabergoline**.
- **Coagulopathy**: no IM injections or spinal/epidural; correct with blood products; deliver vaginally where possible.
- **Gestation 14–28 weeks**: managed on the late-abortion pathway in Uganda with higher misoprostol doses (see also the *Abortion & post-abortion care* topic).
- **Reactive syphilis test**: start benzathine penicillin now and treat the partner.

14. Key Points

High-yield points

- Confirm with **real-time ultrasound**, ideally by a second operator; never on Doppler alone.
- The mother is the patient: check **BP, bleeding, fever and clotting** first.
- DIC: about 10% within 4 weeks, up to about 30% after; worse with abruption or sepsis.
- Urgent delivery for pre-eclampsia, abruption, sepsis, ruptured membranes or DIC.
- **Vaginal birth** for most; CS adds risk now and in every future pregnancy.
- **Bigger baby, smaller dose** of misoprostol; never 200 µg in the third trimester.
- Wait **4 hours** after misoprostol before oxytocin.
- Fresh = intrapartum (labour care); macerated = antepartum cause.
- Examine the **baby and placenta**: the most informative cause-finding step in Uganda.
- Anti-D, cabergoline (not in pre-eclampsia), bereavement care, MPDSR and a 6-week review.

Numbers & doses

Item	Value
Stillbirth threshold (Uganda / WHO comparison)	**28 weeks** or more (ICD: 22 weeks or 500 g)

Item	Value
Fresh vs macerated	Fresh = death within about **12 h** (usually intrapartum); macerated = more than about 12–24 h
Spalding's sign	About **5–7 days** after death
DIC risk	About **10% within 4 weeks**; up to about **30%** after
Bedside clotting test	No clot by **7 minutes** = coagulopathy
Fibrinogen	Normal in pregnancy **4–6 g/L**; **under 2 g/L** = severe coagulopathy
Spontaneous labour if expectant	**85–90% within 2–3 weeks**
Clotting checks if expectant beyond 48 h	**Twice weekly**
Mifepristone	**200 mg orally** once, then misoprostol **24–48 h** later
Misoprostol 14–28 weeks	**400 µg** SL/vaginal every **4–6 h** (WHO) or **200 µg** every 4–6 h (FIGO)
Misoprostol 27–28 weeks	**100 µg** every **4 h** (FIGO)
Misoprostol over 28 weeks	**25 µg** vaginally every **6 h** or **25 µg** orally every **2 h** (FIGO)
25 µg oral solution	1 × 200 µg tablet in **200 mL** water → give **25 mL**
Oxytocin after misoprostol	Wait **at least 4 h**
Oxytocin (WHO MCPC)	2.5 IU in 500 mL, start **10 drops/min**, increase by 10 every **30 min**, max **60 drops/min**
Anti-D (Rh-negative)	Within **72 h**: UK **500 IU** + Kleihauer; the Ugandan vial usually stocked is 1,500 IU
Cabergoline	**1 mg** orally once within **24 h** (not in pre-eclampsia)
Aspirin next pregnancy (placental disease)	**75–150 mg** nightly from **12 weeks**
Follow-up	About **6 weeks**

Exam traps

- **"No FHR on Doppler, so tell her the baby has died."** Wrong: confirm with **real-time ultrasound**, ideally by a second operator.
- **"The fetus is dead, so there is no hurry."** Wrong if she has **pre-eclampsia, abruption, sepsis, ruptured membranes or DIC**: these need urgent delivery.
- **"Do a CS to spare her labour."** Wrong: vaginal birth for most; a scar adds risk in every future pregnancy.
- **"Use 200 µg misoprostol at 36 weeks because the baby is dead anyway."** Wrong: dose **falls** with gestation; 25 µg in the third trimester; hyperstimulation and rupture risk.
- **"Start oxytocin an hour after misoprostol."** Wrong: wait **at least 4 hours**.
- **"Fibrinogen 2.5 g/L is normal."** Wrong in pregnancy: normal is 4–6 g/L, so 2.5 is already falling; under 2 is severe.
- **"Give cabergoline to every woman after stillbirth."** Wrong: **avoid in hypertension/pre-eclampsia**.
- **"Take the Kleihauer after delivery."** Wrong: take it **before delivery** to detect a fetomaternal haemorrhage as the cause.

Questions seniors ask

Viva questions

1. *How do you confirm IUFD?* **Real-time ultrasound showing absent fetal cardiac activity**, ideally confirmed by a second operator.

2. *Why check clotting?* Retained dead fetus → tissue factor release → consumptive coagulopathy. About 10% within 4 weeks and more after. Also abruption and sepsis cause acute DIC.
3. *What is Spalding's sign?* Overlapping of the fetal skull bones on imaging after fetal death (about 5–7 days).
4. *Why do misoprostol doses fall with advancing gestation?* More myometrial receptors and greater uterine sensitivity, so the risk of hyperstimulation and rupture rises.
5. *Fresh vs macerated stillbirth: why does it matter?* Fresh suggests intrapartum death (quality of labour care); macerated suggests an antepartum cause.
6. *When is CS indicated for IUFD?* Rarely: major praevia, uterine rupture, obstructed labour or transverse lie not deliverable vaginally, more than two CS or a classical scar (individualised), severe maternal compromise.
7. *How do you suppress lactation?* Cabergoline 1 mg orally once within 24 hours (not in pre-eclampsia); otherwise supportive measures.
8. *What investigations should every stillbirth in Uganda have at minimum?* BP and urine, FBC and clotting, syphilis and HIV, malaria, glucose, blood group, **careful examination of the baby and the placenta**, and an MPDSR review.

On the ward round

- **How did you confirm the death?** Real-time ultrasound showing no cardiac activity, confirmed by a second operator; Doppler alone is not enough.
- **What four maternal conditions must you exclude at once?** Pre-eclampsia, abruption, sepsis/chorioamnionitis and DIC (plus uterine rupture if scarred).
- **What is your DIC screen?** Platelets, PT/APTT, fibrinogen where available, and a bedside clotting test everywhere; repeat twice weekly if expectant beyond 48 h.
- **Which misoprostol dose for this gestation?** 14–28 weeks: 400 µg (WHO) or 200 µg (FIGO) every 4–6 h; over 28 weeks: 25 µg vaginally 6-hourly or orally 2-hourly; lower with a scar.
- **Fresh or macerated, and why does it matter?** Fresh = intrapartum death (look at labour care and the partograph); macerated = antepartum cause.
- **What investigations would you send to find the cause?** Examination of the baby and placenta with histology, syphilis, HIV, malaria, glucose, Kleihauer (before delivery), bile acids and LFTs if itchy, antiphospholipid antibodies at 6 weeks; postmortem if accepted.
- **What does she need before discharge?** Anti-D if Rh-negative, lactation suppression, contraception discussion, bereavement support, MPDSR notification, and a 6-week appointment.

Mnemonics

Causes of IUFD: "PLACENTA-D"

- **P**re-eclampsia / hypertension
- **L**abour complications (obstructed, ruptured uterus)
- **A**bruption
- **C**ord accidents
- **E**ndocrine (diabetes, thyroid, cholestasis)
- **N**o cause found (unexplained)
- **T**ORCH and other infections (syphilis, malaria, chorioamnionitis)
- **A**nomalies (chromosomal, structural)
- **D**isease of blood (isoimmunisation, APS, sickle cell)

Maternal danger check before induction: "B-B-F-C": **B**P, **B**leeding / abruption, **F**ever, **C**lotting.

Misoprostol by gestation: "Bigger baby, smaller dose"

- 14–28 weeks: 400 (WHO) / 200 (FIGO) µg every 4–6 hours.

- 27–28 weeks: 100 µg.
- Over 28 weeks: 25 µg (every 6 hours vaginal, or every 2 hours oral).

After-birth checklist: "BABIES"

- **B**ereavement care (see, hold, name, memories)
- **A**nti-D
- **B**reasts (cabergoline)
- **I**nvestigations (baby, placenta, bloods)
- **E**xamine and document (fresh vs macerated), **E**MTCT / syphilis treatment
- **S**urveillance (MPDSR) and **S**ix-week follow-up

OSCE tips

> **EXAM FAVOURITE**
>
> In OSCE communication stations ("break the news of IUFD"), examiners look for:
>
> - a **warning shot**;
> - **clear words ("has died")**;
> - **silence and empathy**;
> - **no blame**;
> - a **stepwise plan**;
> - an offer of **seeing and holding the baby**;
> - **checking her understanding**;
> - arranging **follow-up**.

- **Examination station**: perform the obstetric abdominal examination in order (inspection, fundal height, lie, presentation, engagement, auscultation), and say that absent heart sounds are only suggestive and that you would confirm by ultrasound.
- **Prescribing station**: write misoprostol **in micrograms with the route and interval**, matched to gestation and scar status.

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SELF-TEST

Flashcards

26 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Define stillbirth as used in Uganda / WHO international comparison	Baby born with **no signs of life at ≥28 weeks** (or ≥1,000 g). In Uganda, losses <28 weeks are managed as abortions.
2	Gold-standard confirmation of IUFD	**Real-time ultrasound: absent fetal cardiac activity**, ideally confirmed by a **second operator**. Pinard/Doppler alone is not enough.
3	Fresh vs macerated stillbirth	**Fresh**: skin intact – death within ~12 h, usually **intrapartum**. **Macerated**: skin peeling, soft skull – death **before labour** (>12–24 h).
4	What is Spalding's sign?	**Overlapping of fetal skull bones** on imaging after fetal death (~5–7 days).
5	Robert's sign and Deuel's halo sign	**Robert's**: gas in fetal great vessels/heart (~12 h+). **Deuel's halo**: halo around skull from scalp oedema.
6	DIC risk with retained dead fetus (RCOG)	About **10% within 4 weeks** of fetal death, rising to about **30% thereafter**; higher with **abruption or sepsis**.
7	Mechanism of DIC in IUFD	Gradual release of **tissue factor (thromboplastin)** from placenta/decidua → consumption of clotting factors, platelets and **fibrinogen**.
8	Indications for urgent delivery in IUFD	**Pre-eclampsia, abruption, sepsis/chorioamnionitis, ruptured membranes, coagulopathy (DIC)**, uterine rupture (laparotomy).
9	Proportion who labour spontaneously within 2–3 weeks after IUFD	About **85–90%**. If expectant >48 h: check **clotting twice weekly** (RCOG).
10	Preferred mode of delivery after IUFD	**Vaginal birth** for most – CS adds maternal risk now and scar risks in future pregnancies with no fetal benefit.
11	Mifepristone dose before misoprostol induction for IUFD	**Mifepristone 200 mg orally once**, then misoprostol **24–48 hours** later (where available).
12	WHO 2022 regimen for IUFD 14–28 weeks	Mifepristone 200 mg PO, then **misoprostol 400 µg SL or PV every 4–6 hours** (or misoprostol alone 400 µg every 4–6 h).
13	FIGO misoprostol regimens for fetal death by gestation	**13–26 wk: 200 µg every 4–6 h**; **27–28 wk: 100 µg every 4 h**; **>28 wk: 25 µg PV every 6 h or 25 µg oral every 2 h**.
14	Why are misoprostol doses reduced as gestation advances?	Increasing **myometrial receptor density and uterine sensitivity** → higher risk of **hyperstimulation and uterine rupture**.
15	How to give 25 µg misoprostol orally with 200 µg tablets	Dissolve **1 × 200 µg tablet in 200 mL water**; give **25 mL** (= 25 µg).
16	Minimum interval between last misoprostol dose and starting oxytocin	**At least 4 hours**.
17	Lactation suppression after stillbirth	**Cabergoline 1 mg orally, single dose, within 24 h** of birth (avoid in hypertension/pre-eclampsia). Otherwise supportive bra, cold compresses, analgesia. Not bromocriptine.
18	Anti-D after IUFD in an Rh-negative woman	**Anti-D 500 IU IM within 72 h**, with **Kleihauer** (taken before delivery) to detect large FMH and adjust the dose.
19	Single most informative investigation after stillbirth	Examination and **histology of the placenta** (with examination of the baby); postmortem where accepted.

#	Question	Answer
20	Name 6 causes of IUFD common in Uganda	**Pre-eclampsia/hypertension, abruption, malaria, syphilis, intrapartum asphyxia (obstructed labour, uterine rupture), diabetes**, FGR, anomalies, cord accidents.
21	Key elements of breaking news of IUFD (SPIKES)	Private **Setting**; check **Perception**; **Invitation**; **Knowledge** with warning shot and clear words 'has died'; **Empathy**/silence; **Strategy** and follow-up.
22	Four maternal conditions to exclude in the first minutes after suspected IUFD	**Pre-eclampsia** (BP, urine protein), **abruption** (pain, hard uterus, bleeding), **sepsis / chorioamnionitis** (fever, offensive liquor) and **DIC** (oozing, bedside clotting test); plus **uterine rupture** if she has a scar.
23	IUFD: fibrinogen 1.6 g/L, platelets 88, bedside clot not formed at 9 minutes. Interpret and act.	**DIC** (fibrinogen under 2 g/L is severe; normal in pregnancy is 4–6). Senior, fresh whole blood / FFP ± cryoprecipitate and platelets, **deliver vaginally** promptly, oxytocin + TXA for PPH, avoid IM injections and spinal/epidural.
24	Why is Doppler alone not enough to diagnose IUFD?	A **maternal pulse** can be mistaken for the fetal heart, and obesity, polyhydramnios or position can hide a live one. Confirm with **real-time ultrasound (a few minutes on the heart)**, ideally by a **second operator**.
25	Which tests should be taken BEFORE delivery in IUFD?	**Kleihauer–Betke** (fetomaternal haemorrhage), **FBC/platelets, clotting and fibrinogen** (or bedside clotting test), **group and crossmatch**, and **syphilis/HIV/malaria** tests; BP and urine protein.
26	IUFD with severe pre-eclampsia: which lactation-suppression drug should you avoid?	**Cabergoline** (and other ergot-derived dopamine agonists) – avoid in hypertension/pre-eclampsia. Use a firm bra, cold compresses and analgesia instead.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 27-year-old G2P1 at 32 weeks presents to an HC IV with no fetal movements for 24 hours. The midwife cannot hear the fetal heart on a hand-held Doppler. What is the MOST appropriate next step before informing her that the baby has died?

A Tell her now – absent heart sounds on Doppler are diagnostic
B Perform an abdominal X-ray for Spalding's sign
C Confirm with real-time ultrasound, ideally by a second operator
D Repeat the Doppler in 1 week
E Start misoprostol induction immediately

2. A 30-year-old G3P2 at 35 weeks has a confirmed IUFD. BP 168/112, urine protein 3+, headache and epigastric pain. Platelets 88 × 10^9/L. What is the MOST appropriate management?

A Expectant management at home with review in 1 week
B Oral methyldopa and await spontaneous labour
C Immediate caesarean section under spinal anaesthesia
D Misoprostol 400 µg SL every 3 hours and discharge
E $MgSO_4$, antihypertensive therapy and expedited delivery

3. A 26-year-old G1 at 36 weeks with confirmed IUFD is well, normotensive, afebrile, membranes intact, clotting normal. She has no uterine scar. She and her husband want delivery today. Mifepristone is not available. Which regimen is MOST appropriate?

A Misoprostol 400 µg sublingually every 3 hours
B Misoprostol 25 µg vaginally every 6 hours, then oxytocin if needed
C Misoprostol 600 µg orally single dose
D Oxytocin 10 IU IV bolus
E Elective caesarean section

4. A 24-year-old G2P1 at 20 weeks has a confirmed fetal death. She is stable, no scar. Mifepristone and misoprostol are available. Which is the MOST appropriate WHO-recommended regimen?

A Misoprostol 25 µg vaginally every 6 hours
B Oxytocin infusion 10 IU in 1 L
C Mifepristone 600 mg orally only
D Mifepristone 200 mg orally, then after 24–48 h misoprostol 400 µg SL or PV every 4–6 hours
E MVA

5. A 31-year-old at 30 weeks had an IUFD diagnosed 5 weeks ago and declined induction. She now presents with gum bleeding and oozing from a venepuncture site. What is the MOST likely diagnosis?

A Disseminated intravascular coagulation from retained dead fetus
B Immune thrombocytopenia
C Vitamin K deficiency
D Severe malaria
E Heparin overdose

6. A 33-year-old G4P3 at 38 weeks with confirmed IUFD has had one previous lower segment caesarean section. She is well. Which induction approach is MOST appropriate? Advanced

A Misoprostol 200 µg vaginally every 4 hours
B Elective repeat caesarean section is mandatory
C Foley catheter cervical ripening and/or low-dose misoprostol without escalation, then oxytocin, in a hospital with theatre
D Misoprostol 600 µg orally
E Wait for 6 weeks for spontaneous labour without monitoring

7. A primigravida at 39 weeks is admitted in active labour with a good fetal heart rate. After 14 hours with no partograph entries, she delivers a stillborn baby with intact skin and no peeling. What does this MOST strongly suggest?

A Intrapartum death – review quality of labour care
B Antepartum death from syphilis
C Chromosomal abnormality
D Death more than 1 week before delivery
E Maternal diabetes

8. A woman delivers a macerated stillbirth at 34 weeks. Her ANC card shows no syphilis test. What is the MOST appropriate test to add?

A Serum amylase
B Stool microscopy
C Chest X-ray
D Syphilis serology (RPR/rapid test with treponemal confirmation)
E Serum calcium

9. A 28-year-old at 33 weeks presents with sudden constant abdominal pain and dark vaginal bleeding. The uterus is hard and tender, no FHS, and ultrasound confirms IUFD. BP 90/60, pulse 124. Bedside clotting time: no clot at 10 minutes. What is the MOST appropriate management? Advanced

A Expectant management for 2 weeks
B Discharge on iron tablets
C Immediate CS under spinal anaesthesia
D Misoprostol 400 µg every 3 hours
E Resuscitate, transfuse (fresh whole blood/FFP), ARM and oxytocin to expedite vaginal delivery

10. After delivery of a stillborn baby at 37 weeks, a woman with severe pre-eclampsia asks for help to stop her breast milk. What is the MOST appropriate advice? Advanced

A Cabergoline 1 mg orally once
B Bromocriptine 2.5 mg twice daily for 14 days
C Supportive bra, cold compresses, analgesia, and avoiding stimulation; avoid cabergoline
D Express milk every 3 hours
E Oestrogen tablets

11. A Rh-negative woman delivers a stillborn baby at 30 weeks. Which is MOST appropriate?

A No anti-D is needed because the baby is dead
B Anti-D 500 IU IM within 72 hours, with a Kleihauer test to guide additional dosing
C Anti-D only in her next pregnancy
D Anti-D 250 IU orally
E Give Rh-positive blood to neutralise antibodies

12. Parents ask for a caesarean section after being told their baby died at 36 weeks. The mother is healthy, cephalic presentation, no previous uterine surgery. What is the BEST response?

A Explain that vaginal birth is safer for her now and in future pregnancies, discuss induction and analgesia, and support their decision-making
B Agree immediately – it is the parents' wish
C Refuse to discuss further
D Tell them CS is illegal after fetal death
E Offer CS only if they pay privately

13. A 25-year-old G2P1 at 31 weeks has confirmed IUFD. She is well, clotting normal, membranes intact, and wishes to go home for 2 days to arrange family support before induction. What is the MOST appropriate response?

A Refuse – induction must start immediately
B Perform CS the next day
C Allow it with no advice
D Admit and start high-dose misoprostol against her wishes
E Allow it, with advice to return with fever, bleeding, pain or leaking liquor; if delay exceeds 48 h, check clotting twice weekly

14. A woman with a stillbirth at 36 weeks had itching of the palms and soles for 3 weeks before the fetal death. Which investigation is MOST likely to reveal the cause?

A Thyroid function tests
B Antiphospholipid antibodies
C HbA1c
D Serum bile acids and LFTs
E Parvovirus serology

15. After an unexplained stillbirth at 38 weeks, the couple ask how to reduce risk in their next pregnancy. Which advice is MOST appropriate?

A Avoid any pregnancy for 5 years
B Early booking with dating scan, folic acid, screening for diabetes and hypertension, serial growth scans, fetal movement awareness, and planned birth at a CS-capable facility
C Elective CS at 32 weeks
D No special care is needed
E Bed rest from 20 weeks

Answers & explanations

> **1. C** · Confirm with real-time ultrasound, ideally by a second operator
>
> Diagnosis requires **ultrasound showing absent cardiac activity**, ideally confirmed by a second person. **A**: Doppler can miss the heart (obesity, polyhydramnios, position) or pick up the maternal pulse. **B**: Spalding's sign appears only after ~5–7 days and exposes her to radiation unnecessarily. **D**: delays diagnosis and care. **E**: never induce before confirming death.

> **2. E** · $MgSO_4$, antihypertensive therapy and expedited delivery
>
> Severe pre-eclampsia/HELLP → **stabilise with $MgSO_4$ and antihypertensives, then deliver** (vaginal induction preferred for a dead baby unless contraindicated). **A/B**: waiting risks eclampsia, stroke and DIC. **C**: CS is not indicated for a dead fetus and spinal is risky with low platelets. **D**: 400 µg every 3 hours is far too high for 35 weeks, and discharge is dangerous.

3. B · Misoprostol 25 µg vaginally every 6 hours, then oxytocin if needed

In the **third trimester**, use low-dose misoprostol (**25 µg PV 6-hourly** or 25 µg oral 2-hourly), then oxytocin once favourable. **A/C** are first/second-trimester doses – risk of hyperstimulation and uterine rupture at term. **D**: IV bolus oxytocin is dangerous (hypotension) and not an induction method. **E**: CS has no fetal benefit and adds maternal risk.

4. D · Mifepristone 200 mg orally, then after 24–48 h misoprostol 400 µg SL or PV every 4–6 hours

WHO 2022: IUFD 14–28 weeks – **mifepristone 200 mg then misoprostol 400 µg every 4–6 h**. **A**: third-trimester dose – too low, prolonged induction. **C**: wrong dose and mifepristone alone is slow. **B**: few oxytocin receptors at 20 weeks – ineffective. **E**: MVA is not suitable for a 20-week fetus.

5. A · Disseminated intravascular coagulation from retained dead fetus

Retention >4 weeks raises DIC risk (~30%) from **tissue factor release → consumptive coagulopathy**. **B/C/E** are not linked to the history. **D**: malaria can cause thrombocytopenia but the retained dead fetus for 5 weeks is the classic cause.

6. C · Foley catheter cervical ripening and/or low-dose misoprostol without escalation, then oxytocin, in a hospital with theatre

With **one previous LSCS**, vaginal birth is still preferred – use **mechanical ripening (Foley) and/or low-dose misoprostol, careful oxytocin**, close monitoring for rupture, CS-capable facility. **A/D**: high doses at term with a scar risk rupture. **B**: CS is not mandatory and adds another scar. **E**: prolonged retention without clotting checks risks DIC.

7. A · Intrapartum death – review quality of labour care

A **fresh stillbirth** with a documented live fetus at admission = **intrapartum death** – a signal of **quality-of-care gaps** (monitoring, partograph, delays), to be reviewed in MPDSR. **B/C/E** usually cause macerated (antepartum) deaths. **D**: intact skin excludes death days before.

8. D · Syphilis serology (RPR/rapid test with treponemal confirmation)

Syphilis is a common, treatable cause of macerated stillbirth in Uganda – test and treat her and her partner (benzathine penicillin). **A/B/C/E** are not relevant to stillbirth work-up.

9. E · Resuscitate, transfuse (fresh whole blood/FFP), ARM and oxytocin to expedite vaginal delivery

Abruption with shock and DIC: resuscitate, correct coagulopathy (fresh whole blood is often the most available product), **ARM + oxytocin** – vaginal delivery is usually rapid; CS only if maternal condition worsens and delivery isn't imminent. **A/B**: fatal. **C**: spinal is contraindicated with coagulopathy and CS in DIC is dangerous. **D**: high-dose misoprostol risks rupture and isn't the priority.

10. C · Supportive bra, cold compresses, analgesia, and avoiding stimulation; avoid cabergoline

Cabergoline is avoided in hypertension/pre-eclampsia (vasoconstrictive effects). Use **non-pharmacological suppression**. **A**: correct drug otherwise, but contraindicated here. **B**: bromocriptine is not recommended (stroke, MI, hypertension). **D**: stimulates lactation. **E**: oestrogen increases VTE risk and is not recommended.

11. B · Anti-D 500 IU IM within 72 hours, with a Kleihauer test to guide additional dosing

Fetomaternal haemorrhage can sensitise her regardless of fetal death → **anti-D 500 IU IM within 72 h**, **Kleihauer** to detect large FMH. **A/C**: risk isoimmunisation affecting future pregnancies. **D**: anti-D is IM/IV, not oral. **E**: nonsensical and harmful.

12. A · Explain that vaginal birth is safer for her now and in future pregnancies, discuss induction and analgesia, and support their decision-making

Counsel with empathy: **vaginal birth avoids surgical risk and a uterine scar** (rupture/accreta in future pregnancies, crucial in Uganda). Offer good analgesia and support; respect autonomy after informed counselling. **B**: without counselling is poor practice. **C/D/E**: unprofessional or false.

13. E · Allow it, with advice to return with fever, bleeding, pain or leaking liquor; if delay exceeds 48 h, check clotting twice weekly

A short delay is safe if she is well and membranes are intact; give **safety-net advice** and **check clotting twice weekly if >48 h** (RCOG). **A/D** disregard autonomy without medical indication. **C** omits essential safety advice. **B** is not indicated.

14. D · Serum bile acids and LFTs

Pruritus of palms and soles = **intrahepatic cholestasis of pregnancy → bile acids and LFTs**; high bile acids are associated with stillbirth. **A/B/C/E** are part of general work-up but don't fit this history.

15. B · Early booking with dating scan, folic acid, screening for diabetes and hypertension, serial growth scans, fetal movement awareness, and planned birth at a CS-capable facility

Next-pregnancy care: **early booking, preconception folic acid, screening/treating causes, growth surveillance, fetal movement awareness, planned timely birth at a well-equipped facility**. **A**: not evidence-based. **C**: iatrogenic prematurity. **D**: ignores increased recurrence risk and anxiety. **E**: bed rest has no proven benefit and increases VTE.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: Absent fetal movements at 34 weeks

A 29-year-old G3P2 at 34 weeks (early scan) attends Mengo antenatal clinic saying she has not felt the baby move for 2 days. Her ANC card shows 4 visits, an RPR test 'not done – reagents out', HIV negative, Hb 11 g/dL.

1. What history and examination do you do first?

> History: last movements and any earlier reduction; pain/bleeding (abruption); headache/visual/epigastric symptoms (pre-eclampsia); fever/leaking (infection); itching (cholestasis); thirst/diabetes; malaria, IPTp, ITN; trauma; previous CS; blood group. Exam: **BP, pulse, temperature, urine protein**, pallor, bleeding, fundal height, uterine tenderness, **FHS with Doppler**.

2. BP 118/72, urine protein negative, afebrile, fundal height 31 cm, no FHS on Doppler. What next and how do you tell her?

> **Ultrasound** (second operator if possible) confirms no cardiac activity. Break news with **SPIKES**: private room, husband present, warning shot, clear words 'your baby has died', silence, empathy, no blame, explain next steps and that she need not decide everything now.

3. Which investigations do you send?

> **Maternal safety**: FBC/platelets, **clotting/fibrinogen or bedside clotting time**, group & Rh, crossmatch, RFT/LFT. **Cause**: **syphilis (RPR + TPHA/rapid)**, HIV, malaria test, RBS/HbA1c, bile acids if itching, **Kleihauer before delivery** if Rh-negative, urinalysis. Plan: **examine baby and placenta**; placental histology; offer postmortem.

4. She is well, clotting normal, no scar; she wants delivery today. Mifepristone unavailable. Write the induction plan.

> **Misoprostol 25 µg vaginally every 6 hours** (or 25 µg oral 2-hourly – dissolve 200 µg in 200 mL, give 25 mL), stop once in active labour; then **oxytocin** (≥4 h after last misoprostol) if needed, titrated to 3–4 contractions/10 min. **Analgesia (pethidine 50–100 mg IM)**, one-to-one support, private area. Monitor BP, pulse, temp, bleeding. **Active management of third stage** (oxytocin 10 IU IM); prepare for PPH.

5. She delivers a macerated baby with a large, pale placenta. RPR 1:32, TPHA positive. What post-delivery care and follow-up?

> Offer seeing/holding, naming, memories; document maceration and findings; **placental histology**. **Treat syphilis: benzathine penicillin 2.4 MU IM (weekly × 3 if duration unknown) and treat partner**; **cabergoline 1 mg** (normotensive); anti-D if Rh-negative; contraception discussion; mental health support; **MPDSR review** (missed syphilis screening due to stock-out is an avoidable factor); 6-week follow-up and early booking next pregnancy with repeat syphilis screening.

Take-away: Confirm by scan, protect the mother, induce with a gestation-appropriate dose, find the cause – and fix the system gap (here, syphilis testing).

Case 2: IUFD with abruption and DIC

A 35-year-old G5P4 at 33 weeks is referred from a HC III to a regional referral hospital with sudden severe abdominal pain and dark vaginal bleeding for 3 hours. She has had no ANC for 2 months.

1. What are your immediate priorities?

> Call for help; **ABC**, oxygen; **2 large-bore IV lines**; bloods for **Hb, platelets, group & crossmatch (4 units), clotting and bedside clotting test, RFT**; crystalloid bolus; catheter and hourly urine output; **BP and urine protein** (abruption is often associated with pre-eclampsia).

2. Pulse 128, BP 88/56 (booking BP unknown), uterus woody-hard and tender, no FHS; scan confirms IUFD with retroplacental clot. Bedside clotting test: no clot at 10 min. Diagnosis?

> **Placental abruption with IUFD, hypovolaemic shock and DIC** (concealed haemorrhage – visible loss underestimates true loss).

3. Outline management.

> Resuscitate; **transfuse early – fresh whole blood** (or packed cells + FFP, platelets, cryoprecipitate where available); **ARM** then **oxytocin** to expedite vaginal delivery (usually rapid after abruption); analgesia (avoid spinal/epidural); **tranexamic acid 1 g IV** if PPH; CS only if maternal condition deteriorates and delivery is not imminent, after starting correction of coagulopathy. $MgSO_4$ + antihypertensives if pre-eclampsia is found.

4. After delivery she bleeds heavily. What is your approach?

> **PPH bundle**: call for help, uterine massage, **oxytocin 10 IU IM/IV then infusion**, **TXA 1 g IV**, misoprostol 800 µg SL, check placenta complete, examine for tears, **blood products**, bimanual compression/uterine balloon tamponade; escalate to laparotomy (B-Lynch, ligation, hysterectomy) if uncontrolled. Monitor urine output (AKI).

5. What follow-up and prevention advice for the future?

> Counselling and bereavement support; anti-D if Rh-negative; contraception (she is grand multipara – offer long-acting or permanent methods if desired); screen for hypertension postnatally; next pregnancy: early ANC, **aspirin 75–150 mg from 12 weeks** if pre-eclampsia risk, BP monitoring, delivery at a hospital; MPDSR review of delays (no ANC, late referral).

Take-away: In IUFD with abruption the mother is in danger – resuscitate, correct DIC with blood, and deliver vaginally quickly; the dead baby is not the emergency, she is.

Case 3: IUFD at 22 weeks with a previous caesarean

A 31-year-old G3P2 (one SVD, one emergency CS 2 years ago) at 22 weeks is referred to Kawempe National Referral Hospital after a routine scan found no fetal heartbeat. She feels well.

1. How is this loss classified in Uganda and what initial assessment do you do?

> At 22 weeks, Ugandan practice classifies this as a **(missed) late abortion** (<28 weeks), though RCOG would call it a late fetal loss. Confirm by repeat scan; BP, urine, temperature; FBC, clotting, group & Rh, syphilis, HIV, malaria; ask about the CS (type of incision, complications).

2. Mifepristone and misoprostol are available. What regimen do you choose and why?

> **Mifepristone 200 mg PO**, then after 24–48 h **misoprostol** – WHO suggests **400 µg PV/SL every 4–6 hours** for 14–28 weeks, but **with a uterine scar use caution/lower doses (e.g. 200 µg every 4–6 h, FIGO)** and senior oversight. Rationale: mifepristone primes the uterus, reducing total misoprostol needed and hence rupture risk.

3. What will you monitor during induction?

> Pulse, BP, temperature, **scar pain/tenderness**, **vaginal bleeding**, maternal tachycardia/hypotension (rupture signs), contraction frequency (avoid hyperstimulation), progress; count doses; analgesia.

4. She delivers the fetus but the placenta is retained for 90 minutes with moderate bleeding. Next step?

> **Oxytocin 10 IU IM/IV**, empty bladder, controlled cord traction; if not delivered and bleeding, **manual removal or evacuation (MVA/sponge forceps under analgesia/anaesthesia)**; examine placenta; send for histology.

5. What post-delivery care and counselling will you give?

> Anti-D if Rh-negative (250 IU if <20 weeks; 500 IU here as ≥20 weeks); lactation may occur – supportive measures/cabergoline; bereavement care and memories; contraception; investigations results; next pregnancy: early booking, dating scan, aspirin if indicated, scan for placental location (previous CS – accreta risk), plan delivery at hospital.

Take-away: Gestation and a uterine scar decide the prostaglandin dose – mifepristone priming lets you use less misoprostol more safely.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.